

Gabriel Mankowski

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EDUCATION

Iowa State University	Expected May 2027
Bachelor of Science in Computer Engineering (GPA: 3.85)	

SKILLS

- Programming Languages: Python, C, C++, Java, SQL
- Hardware & Tools: Verilog, VHDL, Microchip IGLOO nano, Libero SoC, QuestaSim, Quartus Prime, MSP430, TM4C, RISC-V, Git, Linux
- Cloud & Frameworks: AWS (boto3, S3, Step Functions), FastAPI, Spring Boot, WebSockets

EXPERIENCE

Software Engineer Intern - Innovate!, Inc. (Subcontractor to KBR / USGS TSSC)	May, 2026 - Present
- Engineered asynchronous Python services using FastAPI and AWS SDK (boto3) to track cloud state machine execution and S3 pipeline data transfers for the Annual NLCD. - Developed automated integration and unit test suites to validate data processing workflows across distributed cloud storage environments. - Collaborated with core engineering teams to optimize cloud data retrieval pipelines, decreasing manual monitoring overhead.	

Undergraduate Research Assistant Iowa State University	Jan 2026 – May 2026
- Designed an ultra-low-power ML inference accelerator in VHDL for a Microchip IGLOO nano FPGA using Libero SoC. - Synthesized parallel MAC units, variable quantization (2/4/8-bit), and memory banking to execute LeNet-5 and FastInf models under strict logic cell constraints. - Verified pipeline timing and resolved gate-level propagation delays using QuestaSim/ModelSim simulation testbenches.	

Teaching Assistant – Dept. of Electrical & Computer Engineering	Aug 2025 – Present
- Computer Organization & Assembly (CPRE 3810): Guide students in pipelined RISC-V processor design, hazard mitigation, and cache memory. - Digital Logic (CPRE 2810): Mentored 45+ students weekly on finite state machines, combinational/sequential logic, and Verilog simulation.	

Projects

Obstacle-Avoiding Roomba with IR and Ultrasonic Sensors	Jan, 2025 - May, 2025
- Programmed a TM4C microcontroller to scan environments with IR and PING sensors - Developed algorithms for detecting narrow objects and navigating around obstacles	

ACTIVITIES AND LEADERSHIP

Controlled Environments Club President	Sep 2023 – May 2026
ISU Datathon 3rd place	October 2023
Eagle Scout	